

Hao Bai

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Education

Peking University

Sep. 2023–present

- PhD candidate of Ecology, GPA: 3.82/4.00
- Supervisor: Shengli Tao
- Core courses: Vegetation Ecology, LiDAR Principles and Ecological Practices

Northwest A&F University

Sep. 2019–Jun. 2023

- Bachelor of Forestry, GPA: 3.81/4.00
- Core courses: Forest ecology, Dendrology, Plant Physiology, Forest Management
- Dissertation: Global Estimation of Forest Logging Residues and Associated Greenhouse Effects

Research Experience

Revealing Amazonian Rainforests Damage during the 2023–2024 Droughts

Sep. 2024–Mar. 2026

- Developed a monthly radar satellite dataset (1992–2025) to monitor long-term forest moisture and biomass dynamics across the Amazonian rainforests.
- Quantified forest responses to major historical droughts, with a particular focus on the unprecedented consecutive droughts of 2023–2024.
- Built machine learning models integrating historical observations and CMIP6 climate projections to predict post-drought forest recovery under future climate scenarios.

Global Assessment of Green Roofs' Cooling Potential

Dec. 2023–Feb. 2026

- Compiled a global dataset of ultra-high-resolution building footprints and satellite-derived land surface temperature.
- Developed empirical models relating vegetation cover to daytime and nighttime land surface temperatures across global cities.
- Simulated multiple rooftop greening scenarios to quantify their cooling potential at the global scale.

Selected Publications

1. **Bai, H. et al.** Unprecedented Amazonian rainforests damage during the 2023–2024 droughts. *PNAS*. 123, e2514066123 (2026).
2. **Bai, H. et al.** Cooling potential of global urban roof greening. *npj Urban Sustain* 6, 49 (2026).
3. **Bai, H. et al.** Spaceborne LiDAR reveals 3D structural differences between natural forests and tree plantations in China. *J Plant Ecol* rtag008 (2026)

Awards

President's Scholarship, Peking University (2026)

Outstanding Scientific Research Award, Peking University (2024)

National Scholarship, Ministry of Education, China (2022 & 2020)

Skills

Languages: Chinese (Native), English (IELTS: 7.0)

Programming: Python, MATLAB, R, Google Earth Engine

Field Experience: Conducted terrestrial, backpack, and UAV-based LiDAR surveys in tropical rainforests, temperate forests, and shrublands.